# Chapter 6 Engineering Teacher Key

> Combined teacher answer key + scoring guidance for Chapter 6 (Energy Transformations + Engineering Design). Covers the practice, concept checkpoint, engineering performance task, capstone spec, and student exemplar.

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# ANSWER KEY — Practice (Energy Transformations)

## A. TRANSFER / TRANSFORMATION / BOTH

1. Warm water warms a cool spoon → **TRANSFER** (heat moves; form stays heat)
2. Flashlight (stored → light) → **TRANSFORMATION** (or BOTH if the whole flashlight is traced — accept with reasoning)
3. Toy car hits a cup (cup moves + sound) → **BOTH**
4. Buzzer (electrical → sound) → **BOTH** (electrical moves to the buzzer AND becomes sound)
5. Falling dominoes (motion passes + click) → **BOTH**

## B. INPUT → CHANGE → OUTPUT

1. Wind-up toy: CHANGE = **stored energy → energy of motion** (energy changes form)
2. Flashlight: CHANGE = **stored energy → electrical → light** (energy changes form)
3. Solar panel: INPUT = **light (sunlight)** (light → electrical)

## C. FIX THE ENERGY CHAIN

1. Corrected: **INPUT: stored energy → OUTPUT: sound** (order was reversed).
2. Corrected: **The energy did NOT disappear — it changed form (and a little became sound/heat).**

## D. DESIGN THINKING

1. CRITERIA · 2. CONSTRAINT · 3. CRITERIA · 4. CONSTRAINT

### E. SHORT EXPLANATION

Sample: *A toaster changes electrical energy into heat (and a little motion and sound), which solves the problem of heating bread to just the right temperature.* Accept any device with a correct input → transformation → output that names a solved problem.

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# ANSWER KEY — Concept Checkpoint

## PART 1 — Core Ideas

1. **b) changes form**
2. **b) moves from one object or place to another**
3. **a) transformation**
4. **c) BOTH transfer and transformation**

## PART 2 — Energy Chain

5. CHANGE = **energy changes form (stored energy → energy of motion)** — accept "transformation" or "stored → motion."

## PART 3 — Engineering Design

6. **b) criteria** · 7. **c) constraints** · 8. **b) refine the design**

## PART 4 — Evidence

9. **b) look at the evidence and change ONE thing**
10. Uses **evidence** and **supports** (NOT "proves"). Sample: *Our observation that the bell rang louder is evidence that supports the change helped.*

**Scoring note:** 10 items, 10–15 min; concept score only (not folded into the engineering performance score).

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# ANSWER KEY / SCORING GUIDANCE — Engineering Performance Task

## Page 1 — Problem / Criteria / Constraints

- Problem restated in the team's own words; criteria and constraints listed and addressed.

## Page 2 — Blueprint + Materials

- Materials match the kit; blueprint labels the TRIGGER, OUTPUT, and energy-change location; "how I will know it works" is observable/testable.

## Page 3 — Energy Path

- INPUT → TRANSFORMATION → OUTPUT correct; Transfer/Transformation/BOTH verdict correct AND explained.

## Page 4 — Test 1

- 3 trials recorded; evidence observational (not invented); "what failed" treated as data.

## Page 5 — Redesign

- ONE evidence-based change with a reason + prediction.

## Page 6 — Test 2

- Same conditions as Test 1; Test 1 vs Test 2 compared with evidence.

## Page 7 — Final Explanation

- Claim / Evidence / Reasoning present; uses "supports" (not "proves"); engineering vocabulary used.

## Common correct pathways

- **Bell Swing:** stored (rubber band) → energy of motion (arm) → sound (cup) — BOTH.
- **Flag Drop:** stored (raised object) → energy of motion (falling) → visible signal (flag moves) — BOTH.

- **Domino Signal:** stored (tipped first domino) → energy of motion (transfers down the line) → sound/visible last domino — BOTH.

### What NOT to score for

- Decoration; a failed prototype (score the redesign reasoning); unsupported "heavier = better" claims (require controlled evidence).

### Rubric total

- 15 points (Design 5 · Energy Model 3 · Testing/Redesign 3 · Final Explanation 4).

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## ENGINEERING CAPSTONE — TEACHER SPEC (summary)

- **Problem:** non-electrical tabletop device that signals when a model door/gate/trigger moves; must demonstrate ≥1 energy transformation.
- **Criteria (6):** respond to trigger · observable signal · one transformation · repeated (≥2/3) · explain pathway · supplied materials + tabletop footprint.
- **Constraints (5):** materials only · no electricity/flame/hot plate/launching · time · quantity · footprint.
- **Test protocol:** Test 1 (3 trials) → ONE evidence-based change → Test 2 (3 trials, same conditions) → compare.
- **Safety:** no open flame/hot plate/wall electricity/projectiles; rubber bands stretch only; tabletop only.
- **Data quality:** observational results; failure = data; no invented results.

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## STUDENT EXEMPLAR — "Bell Swing"

- **Design:** craft-stick arm on a string post; a rubber band (stored) releases when a model flap opens; the arm swings a washer into a metal cup → sound.
- **Safety:** tabletop model door (NOT a real door); no heavy swinging bolt; no wall-mounted cowbell.
- **Controlled reasoning:** "We kept the release setup the same and changed only the attached mass. In our test, the signal became louder." Uses **supports/evidence**, NOT "proves."

- **Energy pathway:** stored (rubber band) → energy of motion (arm + washer) → sound (cup) — transfer + transformation = BOTH.
- **Test → Change → Retest:** Test 1 = soft "tick" (1 washer); Test 2 = clear "ding" (+1 washer, same release).
- **Teacher note:** show AFTER students propose designs (modeling/scoring), or to absent students.